Mars Mission Design Project Worksheet

Instructions:

Working in small groups, and as a class, complete each section of the worksheet to design your Mars mission. Use the provided data and any additional resources you deem necessary to support your calculations and decisions.

Use generative AI – Claude or other tools – as a partner in the design process. Ask it to assist with the design of the mission. Keep track of your prompts and questions, as well as its answers.

This is NOT graded. It is complete/not completed only. Upload your answers on Canvas. EACH INDIVIDUAL should upload a completed assignment.

Objective:

Using the mission parameters and constraints provided below, design a roundtrip mission to Mars. Your design Should include calculating a delta-v budget (Delta-V Calculator), planning for crew needs, and selecting a mission trajectory using the NASA Trajectory Browser.

Typical Delta-V Values and Vehicle Information:

Earth to LEO: 9.5 km/s

LEO to Mars Transfer: 3.6 km/s

Mars Orbit Insertion: 1.0 km/s

Mars Surface Ascent: 4.1 km/s

Mars to Earth Return: 2.5 km/s

Propellant: Base calculations on the total delta-v budget, **assuming an average Isp of 450s** for your chemical propulsion engines. Estimate a propellant mass using the rocket equation, considering the mass fractions and engine efficiency given.

MAV (Mars Ascent Vehicle) : 15,000 kg

Life Support and Habitation Module: 20,000 kg

Scientific Instruments and Rovers: 5,000 kg

1. Mission Overview

Objective: Briefly describe the goals of your Mars mission (scientific research, exploration, technology demonstration, etc.). What di you wish to accomplish that requires humans on-site?

Mission Duration: Choose between a minimum energy mission (lowest delta-v) or a fast transit mission (shortest time in flight) and justify your choice.

2. Delta-v Budget Calculation

Using the delta-v values provided above:

- Calculate the total delta-v requirement for your chosen mission profile.
- Include breakdown for each segment: Earth to LEO, LEO to Mars transfer, Mars orbit insertion, Mars surface ascent, Mars to Earth return.
- Discuss the implications of your delta-v budget on the propulsion system and fuel requirements and the flight time.

3. Crew Needs

Describe the life support systems and how you plan to meet the power, communication, and safety requirements for the crew both in flight and on Mars.

4. Mars Surface Operations

Outline the planned activities and objectives for the surface operation. (How long will you be there? See 5 below.) Describe the habitat setup and how the crew can utilize the rovers for exploration.

Discuss the measures taken to protect the crew from radiation and provide thermal control during the mission.

5. Mission Trajectory

Use the NASA Trajectory Browser to select a launch window and plan your mission trajectory. Include a screenshot of your selected trajectory and summarize the key parameters (launch date, arrival date, return date, etc.).

Explain how your chosen trajectory aligns with your delta-v budget and mission objectives.

6. Reflection

Reflect on the challenges you faced during the mission planning process.

Discuss any assumptions or simplifications you made and how they might impact the real-world feasibility of your mission.

Submission:

Compile your calculations, decisions, and trajectory plan into a single doc to upload, Use the submission upload on Canvas. Each student should submit a paper, even if you have worked with others.